

DynamoDB Global Tables – Multi-Region Replication

To Begin with the Lab

Summary of the Lab

In this lab, a DynamoDB table was created in the primary AWS Region and then converted into a Global Table by adding a replica in another Region. The primary table was first created in Mumbai with an order ID partition key, and sample records were inserted to verify replication later. A replica was then added in N. Virginia, which automatically enabled DynamoDB Streams to track item-level changes and keep both tables synchronized. After replication completed, items written in either Region appeared in the other Region as well, showing that both replicas were writable. Finally, the lab confirmed that Global Tables provide multi-region availability, low-latency writes, and continuous data access even if one Region becomes unavailable.

Understanding DynamoDB Global Tables

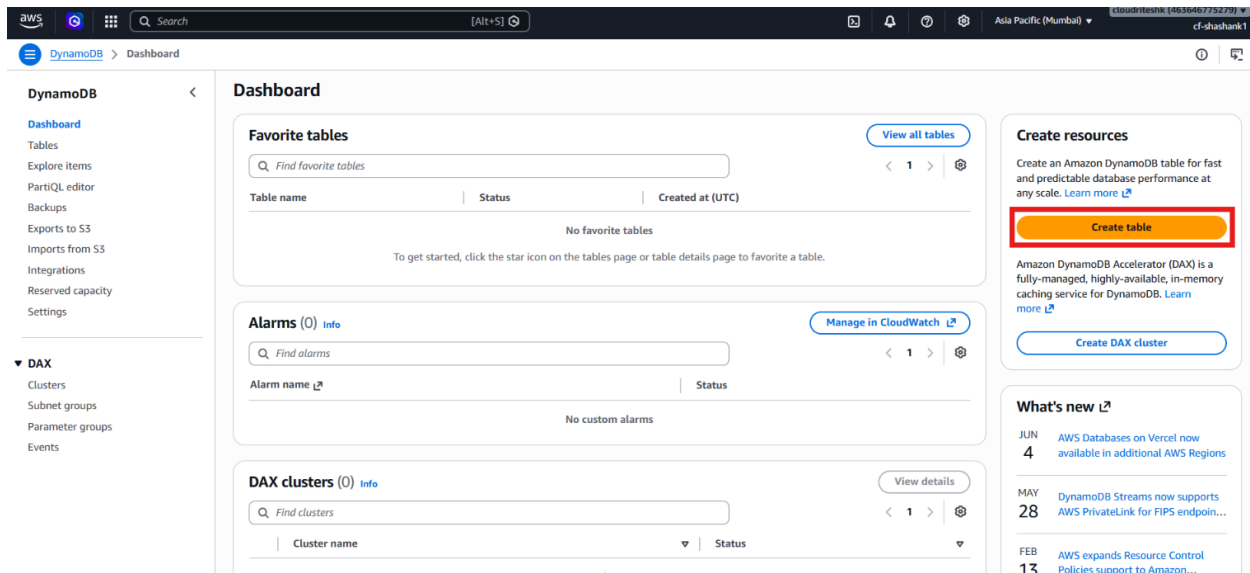
Amazon DynamoDB Global Tables is a multi-region feature that keeps the same table available in more than one AWS Region. It exists so applications can write to the nearest Region, improve response time, and stay available even if one Region has a problem. DynamoDB uses internal replication through DynamoDB Streams to copy item changes between Regions automatically. This gives the application a single logical table with local writes in each Region and fast recovery if one Region goes down.

Example:

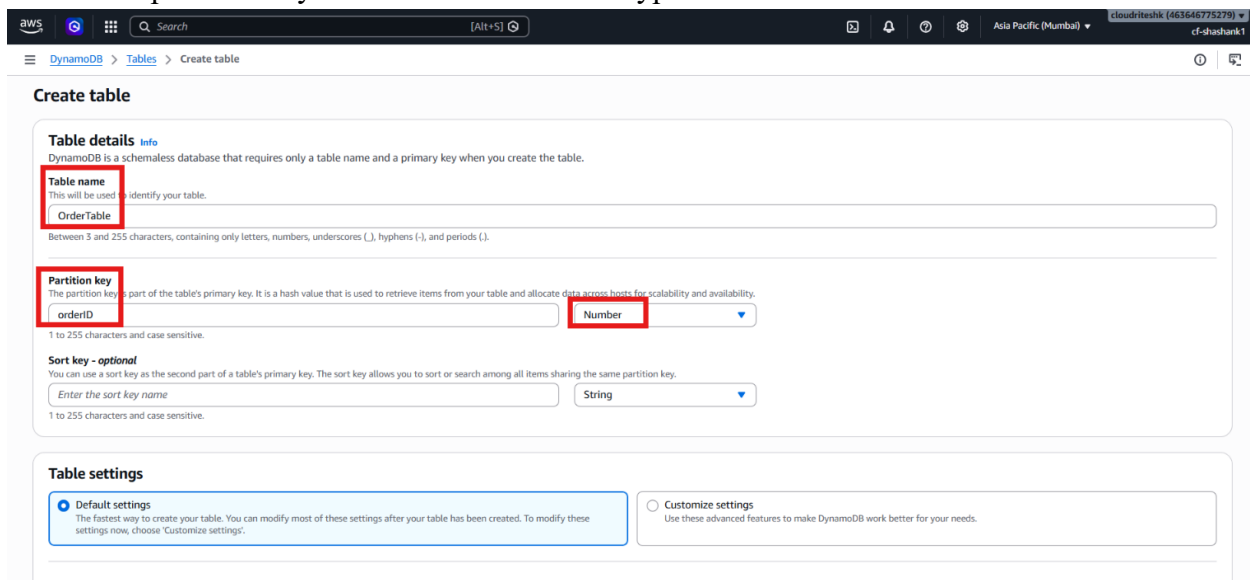
An order table has an item with `orderId = 101` written in Mumbai and another item with `orderId = 103` written in N. Virginia. After replication, both tables show the same records in both Regions, so users in either location see consistent order data. This reduces delay and keeps the application running across Regions.

Lab Steps

- Sign in to the AWS Management Console.
- Open **Amazon DynamoDB** in the Mumbai Region.



- Create a table named **OrderTable**.
- Set the partition key to **orderId** and its data type as **Number**.



- Keep the remaining settings at default values.
- Wait for the table status to become available.
- Click on **Explore table items**.

The screenshot shows the AWS DynamoDB console interface. On the left, the 'DynamoDB' sidebar is visible with options like 'Dashboard', 'Tables', 'Explore items', etc. The main area displays the 'OrderTable' settings. A red box highlights the 'Explore table items' button in the top right corner of the 'OrderTable' section.

- Click on Create item.

The screenshot shows the 'Scan or query items' page for 'OrderTable'. The 'Scan' tab is selected. Below the 'Run' button, a green status bar indicates 'Completed · Items returned: 0 · Items scanned: 0 · Efficiency: 100% · RCUs consumed: 2'. At the bottom, the 'Table: OrderTable - Items returned (0)' section shows a list of items. A red box highlights the 'Create item' button in the top right corner of this section.

- Add an item with orderId = 101.
- Add a string attribute such as name with your chosen value.

The screenshot shows the 'Create item' form. The 'Attributes' section is visible. A red box highlights the 'Add new attribute' button. Below it, the 'Attribute name' field contains 'orderId - Partition key' and the 'Value' field contains '101'. Another red box highlights the 'Name' field, which contains 'Shashank Lodhi', and the 'String' type dropdown. A red box also highlights the 'Create item' button at the bottom right.

- Create the item and verify it appears in the table.

The screenshot shows the AWS DynamoDB console interface. On the left, the 'DynamoDB' sidebar is visible with options like Dashboard, Tables, Explore items, etc. The main area displays the 'OrderTable' details. Under the 'Scan or query items' section, a 'Scan' operation has been completed, returning 1 item. The item details are shown in a table:

orderID (Number)	Name
101	Shashank Lodhi

- Open the table and go to **Exports and streams**.
- Verify that **DynamoDB Streams** is initially off and it will automatically turn on when we will create the **Global table**.

The screenshot shows the 'Exports and streams' tab for the 'OrderTable'. The 'Exports to S3' section shows no exports. The 'DynamoDB stream details' section shows the 'Stream status' as 'Off', which is highlighted in a red box. The 'Turn on' button is visible next to the stream status.

- Open **Global Tables** and click **Create replica**.

The screenshot shows the AWS Management Console interface for a DynamoDB table named 'OrderTable'. The 'Global tables' tab is selected and highlighted with a red box. The 'Settings replication' section indicates that settings replication is disabled. The 'Replicas (0)' section shows that there are no replicas currently configured, and a 'Create replica' button is visible, also highlighted with a red box.

- Select **N. Virginia (us-east-1)** as the replica Region.
- Create the replica.

The screenshot shows the 'Create replica' dialog in the AWS Management Console. The 'Available replication Regions' dropdown menu is open, displaying a list of regions. The region 'United States (N. Virginia) us-east-1' is selected and highlighted with a red box. The 'Create replica' button at the bottom right is also highlighted with a red box.

- Wait for the global table status to finish updating.

Settings replication info

Settings replication determines how table settings and configuration changes are applied across replicas of a global table. Settings replication status can be "Enabled", "Enabled with overrides" or "Disabled".

Settings replication status

Enabled with overrides

Summary

Multi-Region consistency: Eventual

Current Region: Asia Pacific (Mumbai) ap-south-1

Version: 2019.11.21 (Current)

Replicas (2) info

Other AWS Regions to which you have replicated this table.

Replication Region	Status	Type	Capacity mode	Endpoint
United States (N. Virginia) us-east-1	Creating	Replica	On-demand	dynamodb.us-east-1.amazonaws.com
Asia Pacific (Mumbai) ap-south-1	Active	Current Region	On-demand	dynamodb.ap-south-1.amazonaws.com

- Open the table in the N. Virginia Region.
- Verify that the replicated item from Mumbai appears there.

Tables (4) info

Name	Status	Partition key	Sort key	Indexes	Replication Regions	Deletion protection	Favorite	Read capacity mode
Employees	Active	EmpId (N)	-	0 0	Off	☆	Provisioned (5)	
hotelRoomAvailabilityTable	Active	date (S)	-	0 0	Off	☆	On-demand	
hotelRoomBookingTable	Active	bookingID (S)	-	0 0	Off	☆	On-demand	
OrderTable	Active	orderId (N)	-	0 1 Region	Off	☆	On-demand	

- Confirm that **DynamoDB Streams** is now enabled automatically.
- Add another item in the Mumbai table.
- Use a different orderId, such as 102.
- Add a name attribute and save the item.

Create item

You can add, remove, or edit the attributes of an item. You can nest attributes inside other attributes up to 32 levels deep. [Learn more](#)

Attributes

Attribute name	Value	Type	Remove
orderId - Partition key	102	Number	
Name	Tanishq Bansal	String	Remove

Cancel Create item

- Open the N. Virginia replica.

- Verify that the new Mumbai item is replicated there.

The screenshot shows the AWS DynamoDB console interface. On the left, the 'DynamoDB' sidebar is visible with options like 'Dashboard', 'Tables', 'Explore items', etc. The main area displays the 'OrderTable' in the 'United States (N. Virginia)' region. The 'Scan or query items' section shows a completed scan with 2 items returned. The table 'OrderTable' is selected, and the 'Attributes' section shows 'orderID' as the partition key. The scan results show two items: one with orderID 102 and Name 'Tanishq Bansal', and another with orderID 101 and Name 'Shashank Lodhi'.

- Add a new item in the N. Virginia replica.
- Use a different orderID, such as 103.
- Add a name attribute and save the item.

The screenshot shows the 'Create item' form in the AWS DynamoDB console. The 'Attributes' section is visible, showing two attributes: 'orderID - Partition key' with value '103' and 'Name' with value 'Aman Bansal'. The 'Create item' button is highlighted.

- Return to the Mumbai table.
- Refresh and verify that the N. Virginia item is replicated back to Mumbai.

The screenshot shows the AWS DynamoDB console interface. The top navigation bar indicates the current region is 'Asia Pacific (Mumbai)'. The left sidebar shows the 'DynamoDB' section with options like 'Dashboard', 'Tables', 'Explore items', 'PartiQL editor', 'Backups', 'Exports to S3', 'Imports from S3', 'Integrations', 'Reserved capacity', and 'Settings'. The main content area is titled 'OrderTable' and shows a 'Scan' operation. The 'Scan' button is selected, and the 'Table - OrderTable' is chosen. The 'Select attribute projection' is set to 'All attributes'. A green status bar indicates the scan is 'Completed' with 3 items returned, 100% efficiency, and 2 RCUs consumed. Below this, a table titled 'Table: OrderTable - Items returned (3)' shows the scan results. The table has columns for 'orderID (Number)' and 'Name'. The item with orderID 103, 'Aman Bansal', is highlighted with a red box.

orderID (Number)	Name
102	Tanishq Bansal
103	Aman Bansal
101	Shashank Lodhi

- Confirm that both replica tables are writable.
- Confirm that updates are replicated across Regions.
- Confirm that DynamoDB Global Tables provide eventual consistency.
- Confirm that the table remains highly available across multiple AWS Regions.